Duality in LP and KKT conditions

Large Scale Optimization

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1. Duality in LP

Definition of the Dual Problem of an LP

$$\begin{array}{ll} \text{Min}_x & c^\top x \\ s.a: & Ax = b \\ & x \geq 0 \end{array}$$

$\mathbf{(P)}$

$$\begin{array}{lll|l} \text{Min}_x & c_1 \cdot x_1 + ... + c_n \cdot x_n & & \\ s.a: & a_{11} \cdot x_1 + ... + a_{1n} \cdot x_n & = b_1 & \lambda_1 \\ & a_{21} \cdot x_1 + ... + a_{2n} \cdot x_n & = b_2 & \lambda_2 \\ & ... & & \\ & a_{m1} \cdot x_1 + ... + a_{mn} \cdot x_n & = b_m & \lambda_m \\ & x_1 \geq 0, \ ..., x_n \geq 0 & & \end{array}$$

Dual variable $\lambda_i, \ i = 1, ..., m$

$$\begin{array}{ll} \text{Max}_\lambda & b^\top \lambda \\ s.a: & A^\top \lambda \leq c \end{array}$$

$\mathbf{(D)}$

$$\begin{array}{lll} \text{Max}_\lambda & b_1 \cdot \lambda_1 + ... + b_m \cdot \lambda_m & \\ s.a: & a_{11} \cdot \lambda_1 + ... + a_{m1} \cdot \lambda_m & \leq c_1 \\ & a_{12} \cdot \lambda_1 + ... + a_{m2} \cdot \lambda_m & \leq c_2 \\ & ... & \\ & a_{1n} \cdot \lambda_1 + ... + a_{mn} \cdot \lambda_m & \leq c_n \end{array}$$

Duality theorems in LP

Weak theorem

$$\left.\begin{array}{l} x \text{ feasible for (P), } Ax = b,\ x \geq 0, \\ \lambda \text{ feasible for (D), } A^\top \lambda \leq c: \end{array}\right\} \Rightarrow \lambda^\top b = \lambda^\top A x \leq c^\top x$$

Strong theorem

a) (P) has finite solution: $c^\top x^*$, $\Rightarrow$ (D) also: $b^\top \lambda^*$. Moreover:

$$b^\top \lambda^* = c^\top x^*$$

b) (P) is unbounded $\Rightarrow$ (D) is infeasible.

Strong theorem. Proof

Strong theorem

a) (P) has finite solution: $c^\top x^*$, $\Rightarrow$ (D) also: $b^\top \lambda^*$. Moreover:

$$b^\top \lambda^* = c^\top x^*$$

b) (P) is unbounded $\Rightarrow$ (D) is infeasible.

(P) finite: a feasible basic solution exists $x^* = \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix}$,

$r = c_N - N^\top B^{-\top} c_B \geq 0$

$$\left.\begin{array}{l} B^\top u = c_B \\ r = c_N - N^\top B^{-\top} c_B \geq 0 \end{array}\right\} \quad \left.\begin{array}{l} B^\top u = c_B \\ N^\top u + r = c_N \end{array}\right\} \begin{pmatrix} B^\top \\ N^\top \end{pmatrix} u \begin{array}{c} = \\ \leq \end{array} \begin{pmatrix} c_B \\ c_N \end{pmatrix}$$

u is dual feasible: $u^\top b = c_B^\top B^{-1} b \Rightarrow u$ is an optimal solution of (D).

Duality Results. Example

$$
\begin{aligned}
Min \quad & 3x_1 + 4x_2 + 5x_3 \\
& x_1 + 2x_2 + 3x_3 \geq 5 \\
& 2x_1 + 2x_2 + x_3 \geq 6 \\
& x_i \geq 0
\end{aligned}
$$

Optimal Basis and Tableau

1	2	3	4	5	
0	1	5/2	−1	1/2	2
1	0	−2	1	−1	1
0	0	1	1	1	−11

$$I_B = \{2, 1\}$$

$$B = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$$

$$\lambda = B^{-\top} c_B = \begin{pmatrix} 1 & -1 \\ \frac{-1}{2} & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$z_D = b^\top \lambda = (\,5,\ 6\,) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 11$$

Unbounded Primal

$$
\begin{aligned}
Min_x \quad & -x_1 \\
s.a: \quad & x_1 + x_2 - x_3 = 1 \\
& x_i \geq 0
\end{aligned}
$$

$$\Downarrow$$

Infeasible Dual

$$
\begin{aligned}
Max_\lambda \quad & \lambda \\
s.a: \quad & \lambda \leq -1 \\
& \lambda \leq 0 \\
& -\lambda \leq 0
\end{aligned}
$$

Complementary Slackness Theorem

$$\begin{array}{lll} \text{Min}_x & c^\top x & \\ s.a: & Ax = b & \\ \texttt{(P)} & x \geq 0 & \end{array} \qquad \begin{array}{ll} \text{Max}_\lambda & b^\top \lambda \\ s.a: & A^\top \lambda \leq c \\ \texttt{(D)} & \end{array}$$

<u>Strong Duality Theorem:</u> x, λ solve (P) and (D) $\Leftrightarrow c^\top x = b^\top \lambda$

$$\Updownarrow$$

<u>Complementary Slackness Theorem:</u> $\begin{cases} x^\top (c - A^\top \lambda) = 0 \\ c - A^\top \lambda \geq 0 \\ x \geq 0 \\ (Ax = b) \end{cases} \Leftrightarrow x, \lambda$ solve (P) and (

- Assume that $x_{Reord} = \begin{pmatrix} x_B \\ \hdashline x_N \end{pmatrix} = \begin{pmatrix} B^{-1}b \\ \hdashline 0 \end{pmatrix} \geq 0$ solves (P)
- Then, $\lambda = B^{-\top} c_B$ solves (D)

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Infeasible Dual

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\begin{aligned}
Max_\lambda \quad & \lambda \\
s.a: \quad & \lambda \leq -1 \\
& \lambda \leq 0 \\
& -\lambda \leq 0
\end{aligned}
$$

2. KKT conditions in NLP

KKT conditions in NLP

CONSTRAINED NON-LINEAR OPTIMIZATION. Introduction

- **KARUSH-KUHN-TUCKER CONDITIONS.**

  Normal cone at a point of the feasible region.
  1st order necessary conditions and regularity.
  Convex problems. Sufficient conditions.
  Examples

UPC

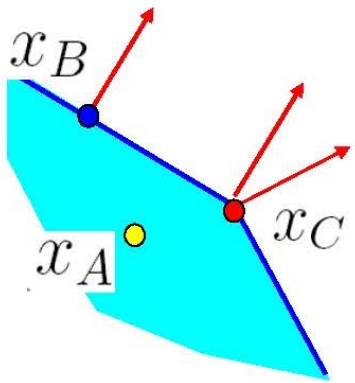
KKT conditions in NLP

$$\begin{array}{ll} Min_x & f(x) \\ s.a: & Ax - b \geq 0 \end{array}$$

$$\bar{x} \in P, \quad N_P(\bar{x}) = \{\, v \in \mathbb{R}^n \,|\, v^\top \Delta x \leq 0, \qquad \bar{x} + \Delta x \in P \,\}$$

$$P = \{\, x \in \mathbb{R}^2 \mid 80 - x_1 - x_2 \geq 0, 100 - 2x_1 - x_2 \geq 0 \,\}$$

$$A = (a_1\ a_2), a_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \ a_2 = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$



$$\begin{aligned} N_P(x_A) = & \ \{0\} \\ N_P(x_B) = & \ \{v = -\mu_1 a_1, \ \mu_1 \geq 0\,\} \\ & \ a_1 x_B - b_1 = 0 \\ N_P(x_C) = & \ \{v = -\mu_1 a_1 - \mu_2 a_2, \ \mu_1, \mu_2 \geq 0\} \\ & \ a_1 x_B - b_1 = 0 \\ & \ a_2 x_B - b_2 = 0 \end{aligned}$$

In general $N_P(x) = \{\, v = -A^\top \mu \,|\, \mu \geq 0, a_i^\top x - b_i > 0 \Rightarrow \mu_i = 0\}$

KKT conditions in NLP

For the problem
$$\begin{array}{ll} Min_x & f(x) \\ s.a: & Ax - b \geq 0 \,\Big|\, \mu \end{array}$$

At a local optimum x^*,

$$\exists \mu \geq 0, \quad \begin{array}{l} \nabla f(x^*) = A^\top \mu \\ (Ax-b)^\top \mu = 0, \\ \mu \geq 0 \end{array} \quad \Rightarrow -\nabla f(x^*) \in N_P(x^*)$$

Assume $-\nabla f(x^*) \notin N_P(x^*)$:

$\exists \Delta x$ t.q.:

a) $x^* + \Delta x^* \in P$

b) $(-\nabla f(x^*))^\top \Delta x^* > 0 \Rightarrow \nabla f(x^*)^\top \Delta x^* < 0$

A descent of obj.f. values occurs along Δx^*

$\Rightarrow x^*$ is not a local minimum

KKT conditions in NLP

A NLP problem of the type:

$$\begin{array}{lll} Min_{\,x} & f(x) & \\ s.a & h(x) = 0 & (P) \\ & g(x) \geq 0 & \end{array}$$

OBJECTIVE:

To study conditions verified by local optima of *(P)*

$$\begin{array}{l} h : R^n \rightarrow R^p, \\ g : R^n \rightarrow R^m, \\ f : R^n \rightarrow R, \end{array} \qquad f, g, h \in \mathcal{C}^1$$

Notation:

$$h(x) = \begin{pmatrix} h_1(x) \\ \vdots \\ h_p(x) \end{pmatrix}, \; g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{pmatrix} \quad \left(\frac{\partial g}{\partial x}\right)_{(x)} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(x) & \dots & \frac{\partial g_1}{\partial x_n}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1}(x) & \dots & \frac{\partial g_m}{\partial x_n}(x) \end{pmatrix}$$

KKT conditions in NLP

$$Min_{\,x} \quad f(x)$$

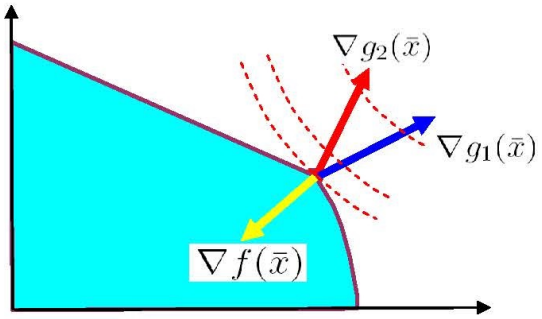
$$s.a \quad h(x) = 0$$
$$g(x) \geq 0$$

$$Min_{\,x} \quad (x_1-3)^2+(x_2-2)^2$$

$$s.a: \quad -g_1(x) = x_1^2+x_2^2 \leq 5$$
$$-g_2(x) = x_1+2x_2 \leq 4$$
$$-g_3(x) = x_1 \leq 0$$
$$-g_4(x) = x_2 \leq 0$$

Index set for active constraints:

$$\mathcal{I}(x) \triangleq \{\, 1 \leq i \leq m \mid g_i(x) = 0 \,\}$$



The set of active constraints at the $\boxed{\text{point} \quad x=(2,1)}$, global minimum of function f, is $\mathcal{I}(x)=\{1,\,2\}$.

$$\nabla f(x) = \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \quad \nabla g_1(x) = \begin{pmatrix} -4 \\ -2 \end{pmatrix}, \quad \nabla g_2(x) = \begin{pmatrix} -1 \\ -2 \end{pmatrix},$$

$$u_0 \begin{pmatrix} -2 \\ -2 \end{pmatrix} = u_1 \begin{pmatrix} -4 \\ -2 \end{pmatrix} + u_2 \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

$$\boxed{\text{Para} \quad u_0 = 1 \;\Rightarrow\; u_1 = 1/3,\; u_2 = 2/3,\; u_3 = u_4 = 0}$$

KKT conditions in NLP

K-K-T NECESSARY CONDITIONS

$$Min_x \quad f(x)$$
$$s.a \quad h(x) = 0$$
$$g(x) \geq 0$$

x^* assumed a local optimum of (P).

Regularity in x^* :

$$J = \frac{\partial(h_1, \ldots, h_p, g_\ell)}{\partial(x_1, \ldots, x_n)}(x), \quad \ell \in \mathcal{I}(x)$$ full rank jacobian

There exist multipliers $u \in \mathbb{R}^m, \; v \in \mathbb{R}^p$ so that:

$$\nabla f(x^*) = \left(\frac{\partial g}{\partial x}\right)^\top_{(x^*)} u + \left(\frac{\partial h}{\partial x}\right)^\top_{(x^*)} v$$

$$u \geq 0, \quad g(x^*) \geq 0, \quad u^\top g(x^*) = 0$$

(u, v unique)

KKT conditions in NLP

K-K-T. SUFFICIENT CONDITIONS

f, g convex , $h = Ax - b$ affine

$$\begin{aligned} Min_x \quad & f(x) \\ s.a \quad & h(x) = 0 \\ & g(x) \geq 0 \end{aligned}$$

If x^* verifies:

$$\nabla f(x^*) = \left(\frac{\partial g}{\partial x} \right)^\top_{(x^*)} u + \left(\frac{\partial h}{\partial x} \right)^\top_{(x^*)} v$$

$$u \geq 0, \quad g(x^*) \geq 0, \quad u^\top g(x^*) = 0$$

then, x^* is a global optimum of the problem

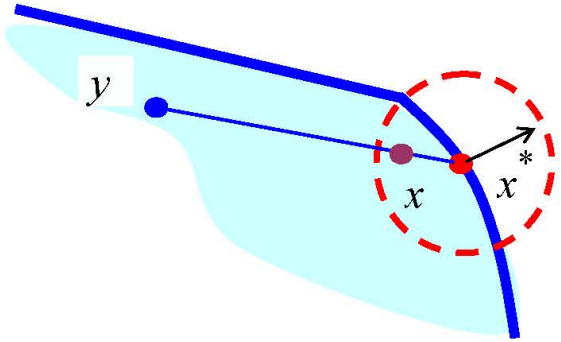
KKT conditions in NLP

Si f convexa sobre $\mathbb{R}^n$ $f(x) \geq f(x_1) + (x - x_1)^\top \nabla f(x_1)$**.**

Igualmente, si f convexa sobre P conjunto convexo:
$f(x) \geq f(x_1) + (x - x_1)^\top \nabla f(x_1),\ \forall\, x, x_1 \in P$

Si x^* es mínimo local de f convexa sobre P convexo $\Rightarrow$
$\Rightarrow$ x^* es mínimo <u>global</u> de f sobre P.

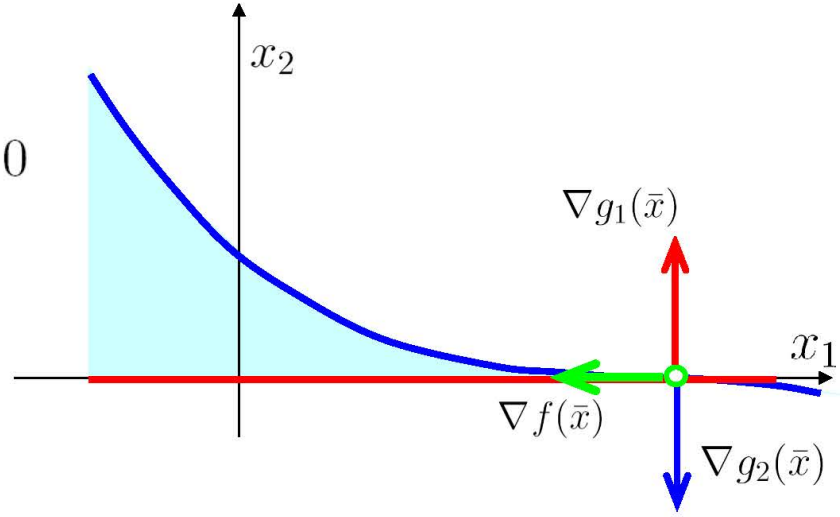
$\nabla f(x^*)^\top (x - x^*) \geq 0,\ \forall x \in B_\delta(x^*) \cap P$
Si $y \in P,\ y \notin B_\delta(x^*) \cap P\ y\bar{x}^* \subset P$
$f(y) - f(x^*) \geq \nabla f(x^*)^\top (y - x^*) = \alpha \nabla f(x^*)^\top (x - x^*) \geq 0$



KKT conditions in NLP

$$Min_{\ x} \quad -x_1$$

$$s.a: \quad g_1(x) = -x_2 + (1-x_1)^3 \geq 0$$
$$g_2(x) = x_2 \geq 0$$



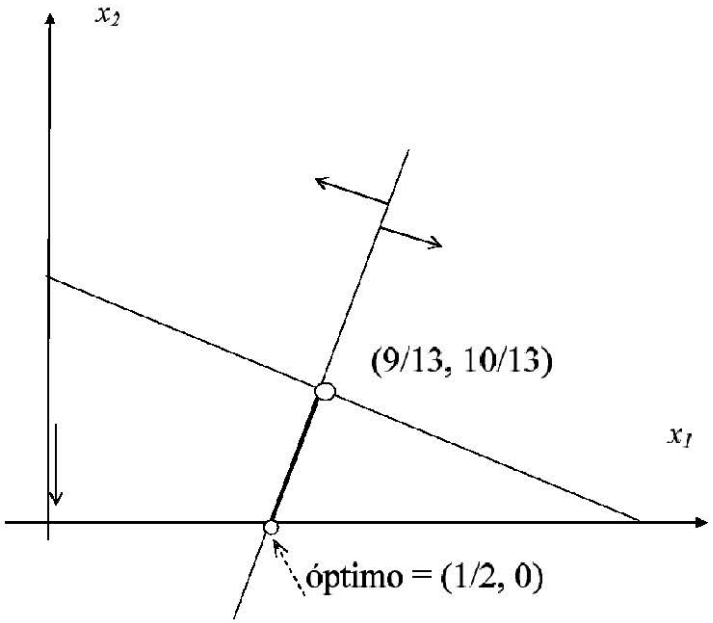
Solución $\bar{x} = (1, 0)$

$$\nabla g_1(x) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \nabla g_2(x) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \nabla f(x) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$u_0 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = u_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + u_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix} \qquad \boxed{\Rightarrow \ u_0 = 0}$$

KKT conditions in NLP

$$
\begin{array}{lll}
Min_x & & x_2 \\
s.a: & -g_1(x) = & x_1 + 3x_2 \leq 3 \\
 & -g_1(x) = & 4x_1 - x_2 \leq 2 \\
 & g_3(x) = & 4x_1 - x_2 \geq 2 \\
 & & x_1, x_2 \geq 0
\end{array}
$$



At optimal points:

$$u_0 \nabla f(x^*) = \left(\frac{\partial g}{\partial x} \right)^\top_{(x^*)} u \qquad u \geq 0, \quad u^\top g(x^*) = 0$$

$$\nabla f(x) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \nabla g_1(x) = \begin{pmatrix} -1 \\ -3 \end{pmatrix}, \quad \nabla g_2(x) = \begin{pmatrix} -4 \\ 1 \end{pmatrix}, \quad \nabla g_3(x) = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$u_0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = u_1 \begin{pmatrix} -1 \\ -3 \end{pmatrix} + u_2 \begin{pmatrix} -4 \\ 1 \end{pmatrix} + u_3 \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

But also at any other point (!!!)

$$u_0 = 0, u_1 = 0, u_2 = u_3 > 0$$

$$x = (9/13, 10/13)$$